

Concrete project ideas for tamper-resistance & open-weight model safety

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This document outlines concrete project ideas I am interested in to improve open-weight model safety by making them more resistant to harmful tampering. See the paper [Open Technical Problems in Open-Weight AI Model Risk Management](#) for additional background and ideas.

This is a public living document of my personal notes & ideas.

This doc might help with the following:

- Helping you know what I'm thinking about.
- If you're working on similar things, helping you coordinate with me so we don't scoop each other.
- Helping you know what I might be able to collaborate on.
- Potentially getting you interested in these things too. I am currently making no attempt to "dibs" any of these ideas.
- Getting errors or gaps in my thinking corrected.

So when it comes to this doc, please choose your own adventure and feel free to forward it to others. Comments are always welcome.

1. When is "[deep ignorance](#)" possible?

This idea is for a project studying what types of tasks models can and cannot be made tamper-resistantly ignorant with respect to.

Hypothesis

There are a lot of potentially harmful behaviors that we might want open-weight models to tamper-resistantly lack. Consider, for example:

- Producing toxic text.
- Producing NSFW content.
- Providing information about a specific pathogen or chemical.
- Providing information related to CBRN weapons.

Emerging evidence from a few different papers has started to lead me to a specific hypothesis about when tamper-resistant ignorance about a topic is and isn't possible. My hypothesis is that *“deep ignorance” and tamper resistance are only possible when the behavior you want resistance to is informationally difficult to learn, conditional on the model knowing everything else it knows*. For example, imagine a human who is generally intelligent but has never seen/experienced toxic speech and who has never learned about the science of biothreats. It might only take them minutes to learn how to say offensive things, while it might take many hours of study to become an expert on biothreats. Based on this anthropomorphic intuition and clues from prior work, I suspect models will be analogous.

Prior work

Here is the evidence I am aware of that we have related to this hypothesis so far. See also this [thread](#).

- **Safety Pretraining: Toward the Next Generation of Safe AI** ([Maini et al., 2025](#)): This paper found in one experiment that models trained without toxic text could be **more** vulnerable to black-box attacks that attempt to elicit toxicity.
- **When Bad Data Leads to Good Models** ([Li et al., 2025](#)): Similar to the above paper, this paper had an experiment showing that models trained without toxic text could be **more** vulnerable to attacks eliciting toxicity.
- **Estimating Worst-Case Frontier Risks of Open-Weight LLMs** ([Wallace et al. 2025](#)): This paper reported an instance where filtering biothreat data didn't have a big impact. But without more info on how and how much they filtered, it is hard to draw clear conclusions.
- **Deep Ignorance: Filtering Pretraining Data Builds Tamper-Resistant Safeguards into Open-Weight LLMs** ([O'Brien et al., 2025](#)): In this paper, we showed that filtering biothreat-related pretraining data is SOTA for making models resist adversarial fine-tuning. See also [Chen et al. \(2025\)](#), which studied data filtration as a safeguard, but did not perform tampering experiments.
- **Beyond Data Filtering: Knowledge Localization for Capability Removal in LLMs** ([Shilov et al., 2025](#)): This paper showed a methodology that improved over simple filtering methods when filter error is present.
- **Best Practices for Biorisk Evaluations on Open-Weight Bio-Foundation Models** ([Wei et al., 2025](#)): This paper tested the filtration of species/genus data against adversarial fine-tuning. It didn't work well. This suggests filtering may work better if applied to entire tasks/domains rather than specific, narrow forms of knowledge.
- **Shaping capabilities with token-level data filtering** ([Rathi et al., 2026](#)) show that token-level filtering can effectively improve resistance to adversarial fine-tuning.
- **Open-weight genome language model safeguards: Assessing robustness via adversarial fine-tuning** ([Black et al., 2025](#)): Related to the above, this paper tests whether filtering specific potentially-risky biological sequences can make models tamper-resistant w.r.t. learning about them. Like the paper above, they found limited success.

- **GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models** ([Nichol et al., 2021](#)): This older (2021) paper trained an image generation model on data that had been filtered for humans. It produced a model organism of human data filtration that is hard to extract human-like features from. But I do not know of much work directly red-teaming it.
- **Evaluating Concept Filtering Defenses against Child Sexual Abuse Material Generation by Text-to-Image Models** ([Cretu et al., 2025](#)): This paper red-teamed image generation models that had been trained on “an ethical proxy for CSAM (a child wearing glasses)” and found that even small amounts of filtering errors could make the model red-teamable for getting images of children wearing glasses.
- **Video Deepfake Abuse: How Company Choices Predictably Shape Misuse Patterns** ([Kamachee et al., 2025](#)): This paper studied empirical usage trends for Stable Diffusion 1.x vs 2.x models. SD 1.x models were trained on LAION, while SD 2.x models were trained on a dataset with pornographic material filtered out. It found dramatically less real-world usage of SD 2.x models for creating NSFW content than SD 1.x models.

Approach

This paper could either study pre-existing model organisms of data filtration and/or train its own models. The key approach would be to see how sample-efficient attempts to learn different filtered capabilities are. One simple but expensive approach would be to train a number of general-purpose text models, each with different content filtered, such as:

- Bio/chem/cyber text;
- Biothreat/chemthreat/cyberoffense text;
- Or text about specific pathogens, chemicals, or hacking techniques;

And then study how tamper-resistant models are w.r.t. general domain knowledge, harmful domain knowledge, and specific harmful knowledge. The key result to look for would be how difficult it is for models to learn harmful information as a function of how broad the category of data that had been filtered from their training data is. For any given task, does significantly broader filtering result in commensurately more tamper resistance?

Implications

This project would help to cement a clearer empirical understanding of what types of capabilities models can be made “deeply ignorant” and “tamper resistant” with respect to. This would have major implications for practical questions for making open models safer. For example:

- Is tamper-resistance w.r.t. some harmful capabilities (e.g., toxicity) even possible using ignorance-based techniques?
- Is it possible for ignorance-based tamper resistance safeguards to defend against misuse narrowly, or does tamper-resistance require ignorance of a broad domain? For example:
 - Can a model be tamper-resistant w.r.t. CSAM if NSFW images of young people are filtered from its data, or would all NSFW content need to be filtered?

- Could models be tamper-resistently bio/chem safe by filtering out information about specific harmful techniques, pathogens, and chemicals, or would we need to more broadly filter bio/chem-threat-related information?

2. Leveraging unlearning algorithms during pretraining

This idea is for a paper that implements unlearning algorithms to suppress harmful capabilities while a model is being pretrained.

Hypothesis

This idea is fairly simple. Data filtration is an example of an effective, but passive technique for making models lack potentially harmful capabilities. On the other hand, unlearning algorithms are characterized by their ability to actively suppress a specific type of capability. This all suggests that using unlearning algorithms during pretraining may be an obvious way to increase the “depth” or “robustness” of a model’s ignorance on a specific topic.

Prior work

Kyle O’Brien and I have talked about this idea since last summer. We called it the “deep aversion” idea in homage to the [“deep ignorance”](#) project. However, we only invested a limited amount of time into these experiments and found some initial difficulty. Due to our training infrastructure for that project (EleutherAI training infra), it was very difficult to modify the training code for models to implement sophisticated unlearning algorithms. We only ever studied gradient ascent and gradient difference methods, which were not promising by our results. However, I still have optimism for this direction because there are much better and more stable unlearning algorithms compared to gradient ascent gradient difference.

Many prior works have attempted to train models in a way that “immunizes” them against or “blocks” learning on a harmful task. These include [Henderson et al. \(2022\)](#), [Zheng et al. \(2023\)](#), [Huang et al. \(2024\)](#), [Sanyal et al. \(2025\)](#), [Anonymous et al. \(2025\)](#), [Liu et al. \(2025\)](#), and [Abdalla et al. \(2025\)](#). However, despite this framing, **none** of these works have used their techniques during pre-training. Like other tamper-resistant unlearning algorithms, they have only implemented solutions during fine-tuning. See section 4.2.1 of [Casper et al. \(2025\)](#) for more related work.

Approach

The approach could build off of prior work such as the “deep ignorance” paper ([O’Brien et al., 2025](#)). However, in addition to just filtering pre-training data, the normal pre-training process could, every N batches, run a step of some unlearning algorithm. There are many options, but I

would gravitate toward using an algorithm that uses both representation-noising (e.g., [Rosati et al., 2024](#)) and entropy-maximization (e.g., [Entesari et al., 2025](#); [Spartalis et al., 2025](#)). See more discussion below in [Section 3](#). For a project like this, there is a high chance of success because the frequency of unlearning batches and the coefficients for the “forget” parts of the loss function can always be tuned to be as small as they need to be to avoid excessive degradation of unrelated capabilities. The main question is just how much these techniques can improve tamper-resistance.

Implications

If successful, this work would improve the state-of-the-art for tamper resistance, which is currently set by pretraining data filtration ([O’Brien et al., 2025](#)) and knowledge localization & ablation ([Shilov et al., 2025](#)). This type of technique would presumably also be very resilient to filtering errors.

3. Improving unlearning algorithms

This project would be about introducing stronger, more tamper-resistant unlearning algorithms to be implemented during post-training (or potentially pre-training, see above in [Section 2](#)).

Note: People mean different things by “unlearning.”¹ But for the purposes of this doc, any algorithm that is designed to suppress some specific capability based on examples of it counts as “unlearning.”

Hypothesis

Currently, “tamper-resistant” post-training algorithms struggle greatly to resist adversarial fine-tuning. Using lessons learned from failures of the past several years, can we design much better post-training algorithms for tamper-resistance?

Prior work

In the past several years, a number of algorithms have been studied for making open models resist malicious tampering. See Section 4.2.1 of [Casper et al. \(2025\)](#). Below, I will attempt a tabular summary of relevant papers. However, it is not exhaustive. Broadly understood, machine unlearning is suppressing some sort of unwanted knowledge/behavior in models. Thousands of papers fit that description. And the “machine unlearning” literature is a mess due in part to a lack of shared benchmarks/infrastructure and discordant definitions about what “machine unlearning” actually means.² So instead of attempting a thorough overview, I will attempt an overview of

¹ Sometimes people use “unlearning” to mean “removing the influence of data” and other times, they use it to mean, “removing a capability of a model.” These are materially different things. But this document uses the latter definition.

² Ibid.

papers that have proposed unlearning algorithms designed to induce tamper resistance. See also section 4.2.2 of [Casper et al. \(2025\)](#).

Paper	Unlearning algorithm	Percieved weaknesses according to Cas
Self-Destructing Models: Increasing the Costs of Harmful Dual Uses of Foundation Models (Henderson et al., 2022)	Training the model to perform the target task and also making several adversarial fine-tuning adversaries fail to extract harmful information.	Only applied to Bert models. This method has not been scaled up (though I would love for it to be).
Self-Destructive Language Model (Wang et al., 2025)	There is a typical utility preservation loss, an unlearning loss, and a final loss term for minimizing the similarity between the benign and adversarial gradients. Then it updates the model to collapse in that direction, but not the current point in parameter space.	Experiments show resistance up to a few thousand examples of training. But I suspect that this method could be red-teamed effectively using reparameterization, gradient clipping, or learning rate warm-up attacks. I would also apply my default expectation. I bet that a good red team could break it much more efficiently than they did.
CTRAP: Embedding Collapse Trap to Safeguard Large Language Models from Harmful Fine-Tuning (Yi et al., 2025)	They train a model that will conditionally “collapse” into repeating the same safe output when it is fine-tuned on malicious data. This method (1) identifies the harmful gradient direction, (2)	Experiments show resistance up to a few thousand examples of training. But I suspect that this method could be red-teamed effectively using reparameterization, gradient clipping, or learning rate warm-up attacks. I would also apply my default expectation. I bet that a good red team could break it much more efficiently than they did.
Vaccine: Perturbation-aware Alignment for Large Language Models against Harmful Fine-tuning Attack (Huang et al., 2024)	A form of latent adversarial training.	This paper reported resistance to a few thousand adversarial fine-tuning steps. But absent external independent red-teaming, I apply my default expectation: that it's not really that strong.
Targeted Vaccine: Safety Alignment for Large Language Models against Harmful Fine-Tuning via Layer-wise Perturbation (Liu et al., 2024)	A form of latent adversarial training, similar to the original Vaccine paper (Huang et al., 2024).	This paper reported resistance to a few thousand adversarial fine-tuning steps. But absent external independent red-teaming, I apply my default expectation: that it's not really that strong.
Booster: Tackling Harmful Fine-tuning for Large Language Models via Attenuating Harmful Perturbation (Huang et al., 2024)	A form of training under a tampering step. It was designed to improve on the “Vaccine” paper (Huang et al., 2024).	This paper reported resistance to a few thousand adversarial fine-tuning steps. But absent external independent red-teaming, I apply my default expectation: that it's not really that strong.
Antibody: Strengthening Defense Against Harmful Fine-Tuning for Large Language Models via Attenuating Harmful Gradient Influence (Anonymous et al., 2025)	A defense method that is focused on making the local loss landscape with respect to harmful fine-tuning flat. There is also another contribution about weighting untrusted fine-tuning datapoints during fine-tuning according to how malicious they might be, but that is not relevant to open models, just fine-tuning APIs.	Even in this paper, flattening the local landscape was not very effective by itself.
AntiDote: Bi-level Adversarial Training for Tamper-Resistant LLMs (Sanyal et al. 2025)	Adversarial unlearning where there is an adversarial hypernetwork generating malicious LoRA weights.	It's not clear how much collapse the hypernetwork exhibited. Also, I might just be missing it, but I can't actually find how many steps they fine-tuned against. I would also apply my default expectation. I bet that a good red team could break it much more efficiently than they did.

Representation Noising: A Defence Mechanism Against Harmful Finetuning (Rosati et al., 2024)	There were three components to the loss. A retain loss, a gradient ascent forget loss, and a noising loss. The noising loss minimized the max mean discrepancy between latents and randomly sampled white noise.	This method was effectively red-teamed by Qi et al. (2024) . It is also known for being somewhat unstable.
Tamper-Resistant Safeguards for Open-Weight LLMs (Tamirisa et al., 2024)	Uses a Rep-e inspired retain loss and trains under an ensemble of fine-tuning attacks with different LRs.	This method was effectively red-teamed by Qi et al. (2024)
On Weaponization-Resistant Large Language Models with Prospect Theoretic Alignment (Cheng et al., 2025)	A form of adversarial training under an adversarial tampering step.	They reported virtually complete robustness to adversarial fine-tuning, at least well past 10k steps. They also tested under different optimizers, LRs, and schedulers. I am extremely skeptical of the results. I bet that this method could be broken with a learning rate warm-up, gradient clipping, or re-parameterization. Absent external independent red-teaming, I apply my default expectation: that it's not really that strong.
Towards LLM Unlearning Resilient to Relearning Attacks: A Sharpness-Aware Minimization Perspective and Beyond (Fan et al., 2025)	Performs sharpness-aware minimization of loss when performing NPO by using small adversarial weight perturbations.	Increased tamper resistance, but only by a few dozen steps of adversarial fine-tuning.
Beyond Superficial Unlearning: Sharpness-Aware Robust Erasure of Hallucinations in Multimodal LLMs (Fang et al., 2026)	Performs sharpness-aware minimization of loss by using small adversarial weight perturbations.	Increased tamper resistance, but only evaluated it under a few dozen steps of relearning.
Improving Alignment and Robustness with Circuit Breakers (Zou et al., 2024)	Train a LoRA Adapter to (1) preserve clean activations and (2) orthogonalize forget activations.	It's simple. But it seems that the forget loss can be optimized for by having the adapter learn to approximate a conditional orthogonal rotation. Also, Che et al. (2024) and O'Brien et al (2025) successfully red-teamed it.
Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs (Sheshadri et al., 2024)	LAT (applied on top of multiple algorithms).	It helped improve tamper resistance, but only by tens of steps. See also O'Brien et al (2025) .
LoX: Low-Rank Extrapolation Robustifies LLM Safety Against Fine-tuning (Perin et al., 2025)	Using singular value decomposition to disentangle safety-related directions from general capability directions in weight matrices.	It's a simple linear approach. And they didn't test it against adversarial fine-tuning. Nor was it designed to defend against it.
Collapse of Irrelevant Representations (CIR) Ensures Robust and Non-Disruptive LLM Unlearning (Sondej and Yang, 2025)	Performs unlearning by only modifying parameters in a way that does not affect the subspace of activations and gradients spanned by the principal components on benign data.	I think some parts of the paper were a little hard for me to understand. Results across the board seem kind of off. Weird behavior from the circuit breaker baseline. Meanwhile, it reports robustness to over 100 epochs of training on WMDP bio forget, which seems implausible. Not even randomly-initialized models are anywhere close to being that resistant.
IMMA: Immunizing text-to-image Models against Malicious Adaptation (Zheng et al., 2023)	This paper was about tamper resistance for image diffusion models. They train the model with a standard 'retain' loss, a 'forget' loss under model tampering.	Abdalla et al (2025) found this algorithm to be fairly weak.
Towards Resilient Safety-driven	This paper was about tamper resistance	They only test resistance up to 500

Unlearning for Diffusion Models against Downstream Fine-tuning (Li et al., 2025)	for image diffusion models. A meta-learning approach that approximates hypergradients using a Moreau envelope approximation.	adversarial training steps. Absent external independent red-teaming, I apply my default expectation: that it's not really that strong
Leveraging Catastrophic Forgetting to Develop Safe Diffusion Models against Malicious Finetuning (Pan et al., 2024)	This paper was about tamper resistance for image diffusion models. "Contrastive learning to increase the latent space distance between clean and harmful data distribution, thereby protecting models from being fine-tuned to generate harmful images due to forgetting." Essentially, it trains the model to make its latents for harmful data far away from other data using the guidance of a randomly sampled matrix.	This paper only demonstrates resistance to a hundred fine-tuning steps. Also, using a random matrix as a guide seems to indicate that this will not induce terribly complex transformations to how the 'forget' samples are represented.
GIFT: Gradient-aware Immunization of diffusion models against malicious Fine-Tuning with safe concepts retention (Abdalla et al., 2025)	This paper was about tamper resistance for image diffusion models. They train the model with a standard 'retain' loss, a 'forget' loss to minimize performance on the harmful task, a representation noising loss inspired by Rosati et al. (2024), and a local loss landscape curvature loss.	This method might be decent. It seemed to be effective in their experiments (up to several thousand fine-tuning steps), but it has not independently been red-teamed. I also wish that they had much more thorough evals than the example-based ones they presented. But unless proven otherwise, I would apply my default expectation. I bet that a good red team could break it much more efficiently than they did.

As reflected in the table, my experience has led me to have skepticism about the effectiveness of past methods. Absent convincing resistance to second-party red-teaming, my default expectation for results reporting resistance to thousands of steps of adversarial fine-tuning is disbelief – even if this is reported across dozens of fine-tuning configurations. Ultimately, while research on defenses often reports model resistance to thousands or tens of thousands of examples of adversarial fine-tuning, to the best of our knowledge, the state of the art for tamper-resistance, as assessed by second-party red-teaming efforts, is only around several hundred steps of adversarial fine-tuning (e.g., [Qi et al., 2024](#); [Lucki et al., 2024](#); [Che et al., 2025](#)). Meanwhile, to my knowledge, no post-training tamper-resistance algorithms are currently known to make models fully robust to fine-tuning on non-adversarial data on benign topics (e.g., [Hu et al., 2024](#); [Deeb and Roger et al., 2024](#)).

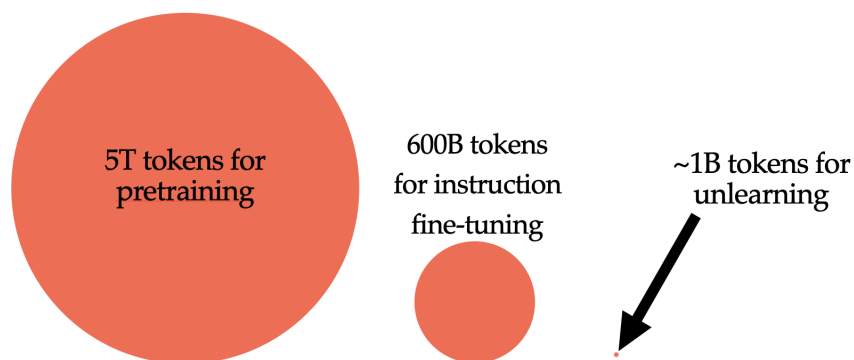
Approaches

First and foremost, given the fraught nature of the machine unlearning field, I think that additional progress will best be made using (1) preregistration of experiments, and (2) adversarial collaborations between dedicated blue team and red team members.

Next, on the blue-teaming side, several innovations may help to improve the tamper resistance of post-training algorithms for open-weight model safety. Here, I outline the idea and rationale for five possible approaches that could be studied either individually or in combination with each other.

Running the algorithms for longer

Consider how we train models. We can look at the numbers from OLMo2 (because they are transparently reported). OLMo2 was pretrained on 5 trillion tokens of web text and instruction-fine-tuned on 600 billion tokens. Compare that to a typical unlearning intervention to suppress harmful capabilities such as knowledge of CBRN weapons. And 1 billion tokens is being a bit generous. For example, the WMDP-bio-forget dataset is only about 150M tokens.



Maybe unlearning algorithms persistently struggle to install tamper-resistant safeguards simply because they don't work on the model for long enough. The corresponding idea is simple: what if we ran tamper-resistant unlearning algorithms for longer, potentially 10 or 100 times longer? Would that make a difference? Avoiding the degradation of harmless capabilities would be simple – just intersperse the unlearning with normal fine-tuning on harmless data.

Importantly, I DON'T think this would work for some unlearning techniques (e.g., [Pal et al., 2025](#)). I think it would only work if the 'forget' objective is genuinely hard to learn and cannot be minimized with a simple solution. For example, take Circuit Breaking ([Zou et al., 2024](#)). It is known not to be very tamper-resistant, and notably, the authors who introduced it did not suggest that it was! But it offers a useful example. Circuit Breaking works by training LoRA adapters to (1) preserve neural activations on 'retain' data and (2) make neural activations on 'forget' data orthogonal to how they previously were. That's it. If you were a LoRA adapter, how would you optimize for this objective? I would simply learn to approximate a simple conditional orthogonal rotation for the 'forget' data. A simple transformation like this should be expected to be removed easily by further fine-tuning. Instead, in order to take advantage of scale with unlearning, we may need algorithms whose objective functions are not so easy to satisfy. And as a corollary, I think that a characteristic of 'good' unlearning algorithms (ones which are better than a simple bag of words classifier) will be that they benefit from scaling.

Fine-tune the full model, not a low-rank adapter

Many past works have aimed to perform tamper-resistant unlearning via LoRA fine-tuning (e.g., [Zou et al., 2024](#); [Sheshadri et al., 2024](#); [O'Brien et al., 2025](#)). But LoRA learns less and forgets less ([Biderman et al., 2024](#)). Similarly to how I speculated a few lines above that 'good' unlearning algorithms should be expected to improve with data scaling, I also expect that 'good' unlearning algorithms will be ones that are better implemented by training the full parameter set.

Running the algorithms on better data

Unlearning algorithms can sometimes be unstable and result in off-target effects. One reason for this is that the ‘forget’ data, while it focuses on the unlearning topic, is full of natural documents that contain a lot of text with a lot of banal, benign text mixed in. As such, compressed or even fully synthetic ‘forget’ datasets may offer a much more incisive way to perform unlearning.

Running the algorithms in ways that can improve neuroplasticity

Just because a model is undergoing training doesn’t mean that it is changing much internally. Some techniques (e.g., [Zunino et al., 2018](#)) have been researched as ways of improving the model’s neuroplasticity. See [Bakhshan et al. \(2026\)](#). However, note that I have not done much work to review lit on techniques like this. For what it’s worth, I’ll share one hunch that I recently had about how training models using a carousel of different optimizers and parameterizations might be an interesting way to have the model updates “access” the capabilities one wants to remove.

Knowledge corruption & synthetic document training

It is possible that the persistent struggles of post-training unlearning methods are due, in part, to how unlearning methods do not typically try to replace unlearned knowledge with anything. Relying on ignorance alone may fail to give the network competing information to learn, thus making it difficult to fully unlearn what may already have been there. There is an emerging literature on “knowledge corruption” and “synthetic document” training in state-of-the-art LLMs (e.g., [Wang et al., 2025](#); [Slocum et al., 2025](#)) and some challenges with naive approaches ([O’Brien et al., 2025](#)). Currently, it seems very possible that tamper-resistant unlearning approaches could be augmented by teaching models harmless alternative information about risky topics. There seems to be low-hanging fruit here.

Placeholder: TODO Cas write about my idea for knowledge corruption where we write a “textbook”, corrupt it coherently, and then use that to rewrite the corpus.

Improving ‘forget’ losses to better destroy latent information

Some approaches to tamper-resistant unlearning involve adapting the model such that previously learned information will be destroyed from its activations on the ‘forget’ data. But these approaches, as have been implemented in the past, have probably not been well-equipped to succeed.

- [Rosati et al. \(2024\)](#) train the model activations on the ‘forget’ set to approximate random noise as measured by Max Mean Distance (across a batch) with randomly-sampled white noise. A similar approach is used in [Abdalla et al. \(2025\)](#).
 - Unfortunately, having a Gaussian shape does not imply being genuinely random. The way that activations are optimized to resemble random noise in this case will probably fail to reliably scrub away information. Instead, it will probably just contract the distribution of latents to be the shape of a Gaussian.

- [Zou et al. \(2024\)](#) train the model activations on the ‘forget’ set to be orthogonal to how they once were.
 - Unfortunately, this could be cheaply fit by approximating a conditional orthogonal transformation of the ‘forget’ activations – a very cheap transformation that does not destroy information well.
- [Pan et al. \(2024\)](#) train the model activations on the ‘forget’ set to be far from where they once were, using the guidance of a random matrix.
 - Unfortunately, this also seems to be a fairly cheap transformation that does not destroy information well.

Given these past approaches and their struggles, it may be useful to design loss functions that are more principled and cannot be easily optimized for by learning simple translations, rotations, contractions, etc.

- **Making the activations on harmful examples approximate random noise in a way that is genuinely hard to fit:** One simple way to do this might be to modify the ‘forget’ loss from [Zou et al. \(2024\)](#), which merely requires new activations to be orthogonal to the old ones for harmful examples. Instead, we could also optimize the new activations for harmful examples to be orthogonal to each other *across each batch*. This would add additional optimization pressure to transform different activations from the examples in the ‘forget’ set in different ways.
- **Adding in an entropy maximization loss:** For good measure, we could also train the model to output more uniform sampling probabilities across its outputs. See [Entesari et al. \(2025\)](#) and [Spartalis et al. \(2025\)](#), who used entropy maximization techniques but did not test tamper resistance, for inspiration.
- **Using an adversarial setup with nonlinear probes:** We could train the model’s latent activations to lack task-relevant information that can be extracted by an ensemble of nonlinear probes. See [Sharma et al. \(2024\)](#), [Xuan et al. \(2025\)](#), and [Han et al. \(2025\)](#) for inspiration.

Adversarial training under *diverse* tampering threats.

Many approaches to tamper-resistant unlearning have used adversarial training under model tampering attacks. (e.g., [Henderson et al., 2022](#); [Huang et al., 2024](#); [Huang et al., 2024](#); [Sheshadri et al., 2024](#); [O’Brien et al., 2025](#); [Tamirisa et al., 2024](#); [Fan et al., 2025](#); [Sanyal et al., 2025](#); [Cheng et al., 2025](#)). These prior works each train against a particular class of adversarial attacks, such as residual-stream perturbations and weight-space perturbations. These techniques improve tamper resistance, but not by much. What if we turned up adversarial training to an 11?

In the process of eliciting harmful capabilities from a model using adversarial attacks, there are many decisions that can be made:

- Parameterization: Should I attack the model’s inputs, latents, or weights?
- Optimization: How should I optimize my adversarial attack?
- Scope: Should my attacks be example-specific or universal?

Instead of training against a single class of attacks, what if we adversarially trained models under a diverse, but strategically-selected suite of differently configured input-space (e.g., [Hackett et al., 2025](#); [Gulyamov et al., 2025](#)), embedding-space (e.g., [Xhonneux et al., 2024](#)), latent-space (e.g., [Sheshadri et al., 2024](#); [Bailey et al., 2024](#)), discriminative probing (e.g., [Sharma et al., 2024](#); [Xuan et al., 2025](#); [Han et al., 2025](#)), LoRA ([Lermen et al., 2023](#); [Sanyal et al., 2025](#)), and full-parameter (e.g., [Tamirisa et al., 2024](#)) adversarial attacks? Some inspiration can be taken from [Dekany et al. \(2025\)](#), who fine-tuned models under input and embedding space attacks to make them more robust. If that worked, why not try the same approach on steroids? State-of-the-art defenses against input-space prompt injection attacks involve training under diverse classes of input-space adversaries. So why would tampering attacks be any different?

Implications

If successful at significantly improving tamper-resistant post-training unlearning methods, these techniques could improve the open-weight model safety toolbox. And they do not need to be competitive with pretraining interventions to be useful. First, post-training algorithms are typically much cheaper to implement than pre-training interventions. But more importantly, these methods are not at all mutually exclusive with pretraining methods (or each other). They could be used either during (see [Section 2](#)) or in addition to pre-training techniques to achieve SOTA robust unlearning.